

Statistical Code Sample: Reproducible Survival Analysis of TCGA Breast Cancer Data in R

Applicant: Henry Serpa

Project: TCGA Breast Cancer Survival Dashboard

Software: R, survival, survminer, tidyverse

Dataset: TCGA Pan-Cancer Clinical Data Resource

Short Project Report

Background

Cancer survival varies by clinical characteristics such as age and tumor stage. This project uses open TCGA clinical data to demonstrate survival analysis methods.

Objective

To evaluate whether age and pathological stage are associated with overall survival among TCGA-BRCA breast cancer patients.

Methods

The TCGA-CDR clinical dataset was filtered to BRCA patients. Overall survival was defined using OS as the event indicator and *OS.time* as follow-up time. Patients with missing survival outcome, missing follow-up time, or non-positive follow-up time were excluded from the analysis-ready cohort. Kaplan-Meier curves were estimated by pathological stage. Cox proportional hazards regression was used to estimate associations between age, stage, and mortality hazard. The proportional hazards assumption was assessed using Schoenfeld residuals.

Results

The final cohort included 1,083 patients and 151 overall survival events. Median follow-up was 28.3 months and median age was 58 years. The Kaplan-Meier analysis showed lower survival among patients with higher stage disease. In the Cox model, each additional year of age was associated with increased mortality hazard (HR 1.04, 95% CI 1.02-1.05). Compared with Stage I, hazard ratios were 1.80 for Stage II, 3.61 for Stage III, and 12.1 for Stage IV.

Model diagnostics

The Schoenfeld residual test suggested possible non-proportionality for stage group and the global Cox model. Therefore, stage-related hazard ratios should be interpreted as average associations over follow-up.

Limitations

TCGA is not population-based, and follow-up may be limited. The analysis is intended as a reproducible portfolio demonstration rather than a clinical decision-support model.

Conclusion

This project demonstrates reproducible data cleaning, survival analysis, Cox regression, diagnostic checking, and visualization of cancer epidemiology results.

Code Sample

1. Inspect Excel Sheets

Purpose: inspect the TCGA-CDR Excel workbook, identify available sheets, and confirm which sheet contains the patient-level clinical and survival data.

```
library(readxl)
library(tidyverse)
library(here)

file_path <- here("data", "raw", "TCGA-CDR-SupplementalTableS1.xlsx")

# List all sheet names in the workbook before reading any data.
# The CDR file contains multiple sheets; we need to confirm the correct one
# ("TCGA-CDR") before subsetting in the cleaning script.
sheets <- excel_sheets(file_path)
print(sheets)

# Preview the first five rows of each sheet to check column names and
# data types without loading the full file into memory.
for (s in sheets) {
  cat("\n\n--- Sheet:", s, "---\n")
  temp <- read_excel(file_path, sheet = s, n_max = 5)
  print(names(temp))
}
# Result: the "TCGA-CDR" sheet contains the patient-level clinical and
# survival outcome data used in all subsequent scripts.
```

2. Clean TCGA-CDR Data

Purpose: filter to TCGA-BRCA patients, clean age/stage/survival variables, and export an analysis-ready dataset.

```
# =====
# 02_clean_tcg_cdr.R
# Data management and cleaning – TCGA-BRCA cohort
#
# Purpose:
#   Read the raw TCGA Pan-Cancer Clinical Data Resource (TCGA-CDR) Excel file,
#   subset to breast cancer patients (TCGA-BRCA), recode and clean the
#   variables needed for survival analysis, and write an analysis-ready CSV.
#
# Input:  data/raw/TCGA-CDR-SupplementalTableS1.xlsx
# Output: data/processed/tcg_brca_survival_clean.csv
# =====

library(readxl)
library(tidyverse)
library(janitor)
library(here)

file_path <- here("data", "raw", "TCGA-CDR-SupplementalTableS1.xlsx")

# Read the CDR sheet and standardise all column names to snake_case
raw <- read_excel(file_path, sheet = "TCGA-CDR") |>
  clean_names()

# Inspect available cancer types before subsetting
print(sort(unique(raw$type)))

# -----
# Subset to TCGA-BRCA and select/recode analysis variables
# -----
# suppressWarnings() is used for as.numeric() coercion because some cells
# contain "#N/A" strings exported from Excel, which produce expected NAs.
# These are also handled explicitly with na_if() for character columns.

clean_brca <- raw |>
  filter(type == "BRCA") |>
  transmute(
    patient_id = bcr_patient_barcode,
    cancer_type = type,

    # Continuous age; coerce to numeric (Excel sometimes stores as character)
    age = suppressWarnings(as.numeric(age_at_initial_pathologic_diagnosis)),

    # Replace "#N/A" strings (Excel artefact) with proper NA
    gender = na_if(gender, "#N/A"),
    race = na_if(race, "#N/A"),
    stage_raw = na_if(ajcc_pathologic_tumor_stage, "#N/A"),

    # Overall survival: 1 = death, 0 = censored (alive at last follow-up)
    os_event = suppressWarnings(as.numeric(os)),
    os_time_days = suppressWarnings(as.numeric(os_time)),

    # Convert follow-up to months; 30.44 = average days per calendar month
    os_time_months = os_time_days / 30.44
```

```

) |>
mutate(
  # Collapse detailed AJCC sub-stages (e.g. "Stage IIA", "Stage IIIC") into
  # four main ordinal groups used in the survival model.
  # str_detect() matches the most specific pattern first (IV before I).
  stage_group = case_when(
    str_detect(stage_raw, "Stage IV") ~ "Stage IV",
    str_detect(stage_raw, "Stage III") ~ "Stage III",
    str_detect(stage_raw, "Stage II") ~ "Stage II",
    str_detect(stage_raw, "Stage I") ~ "Stage I",
    TRUE ~ NA_character_
  ),

  # Clinically meaningful age bands for descriptive tables
  age_group = case_when(
    age < 50 ~ "<50",
    age >= 50 & age < 65 ~ "50-64",
    age >= 65 ~ "65+",
    TRUE ~ NA_character_
  )
) |>
# Exclude patients with missing survival outcome or zero/negative follow-up.
# Zero follow-up causes undefined log(time) in survival models.
filter(
  !is.na(os_event),
  !is.na(os_time_days),
  os_time_days > 0
)

# -----
# Quick data quality checks
# -----
cat("\nNumber of BRCA patients:", nrow(clean_brca), "\n")
cat("Number of OS events:", sum(clean_brca$os_event, na.rm = TRUE), "\n")
cat("Median follow-up in months:", median(clean_brca$os_time_months, na.rm = TRUE), "\n\n")

cat("Stage distribution (including NA):\n")
print(table(clean_brca$stage_group, useNA = "ifany"))

cat("\nGender distribution:\n")
print(table(clean_brca$gender, useNA = "ifany"))

# -----
# Export
# -----
write_csv(
  clean_brca,
  here("data", "processed", "tcga_brca_survival_clean.csv")
)

cat("\nCleaned BRCA survival dataset saved to data/processed/tcga_brca_survival_clean.csv\n")

```

3. Descriptive Analysis

Purpose: summarize cohort size, survival events, follow-up time, age, gender, and pathological stage distribution. These summaries feed directly into the Streamlit dashboard and inform the analytical choices in scripts 04 and 05.

```
library(tidyverse)
library(here)

df <- read_csv(
  here("data", "processed", "tcga_brca_survival_clean.csv"),
  show_col_types = FALSE
)

# Top-line cohort metrics reported in the project summary
cohort_summary <- tibble(
  metric = c(
    "Number of patients",
    "Number of OS events",
    "OS event proportion",
    "Median follow-up, months",
    "Median age at diagnosis",
    "Patients with missing stage"
  ),
  value = c(
    nrow(df),
    sum(df$os_event, na.rm = TRUE),
    round(mean(df$os_event, na.rm = TRUE), 3),
    round(median(df$os_time_months, na.rm = TRUE), 1),
    round(median(df$age, na.rm = TRUE), 1),
    sum(is.na(df$stage_group))
  )
)

# Stage distribution – used to check whether Stage IV is too sparse for
# reliable Cox estimates (n = 20 in this cohort, noted as a limitation)
stage_summary <- df |>
  count(stage_group, name = "n") |>
  mutate(
    percent = round(100 * n / sum(n), 1)
  )

# Gender and age-group summaries for cohort characterisation table
gender_summary <- df |>
  count(gender, name = "n") |>
  mutate(
    percent = round(100 * n / sum(n), 1)
  )

age_summary <- df |>
  count(age_group, name = "n") |>
  mutate(
    percent = round(100 * n / sum(n), 1)
  )

write_csv(cohort_summary, here("data", "processed", "cohort_summary.csv"))
write_csv(stage_summary, here("data", "processed", "stage_summary.csv"))
write_csv(gender_summary, here("data", "processed", "gender_summary.csv"))
write_csv(age_summary, here("data", "processed", "age_summary.csv"))
```

```
cat("\nCohort summary:\n")
print(cohort_summary)

cat("\nStage summary:\n")
print(stage_summary)

cat("\nGender summary:\n")
print(gender_summary)

cat("\nAge group summary:\n")
print(age_summary)

cat("\nDescriptive summary files saved to data/processed/\n")
# Result: the cleaned cohort includes 1,083 patients and 151 overall survival
# events (event proportion 13.9%). Median follow-up was 28.3 months and median
# age at diagnosis was 58 years. Stage II was the most common stage (56.5%),
# followed by Stage III (22.6%) and Stage I (16.8%). Stage IV accounted for
# only 1.8% of the cohort (n = 20), a small subgroup flagged as a limitation
# in the Cox model interpretation. The cohort was 98.9% female, as expected
# for TCGA-BRCA. Twenty-four patients (2.2%) had missing stage information
# and are excluded from the main Cox model.
```

4. Kaplan-Meier Survival Analysis

Purpose: estimate overall survival curves by pathological stage and export a publication-style survival plot.

```
# =====
# 04_kaplan_meier.R
# Kaplan-Meier survival estimation and log-rank test – TCGA-BRCA
#
# Purpose:
#   Estimate overall survival curves stratified by pathological tumor stage
#   using the Kaplan-Meier (product-limit) estimator. A log-rank test is
#   used to evaluate whether survival distributions differ across stage groups.
#   The output figure is used in the Streamlit dashboard.
#
# Input:  data/processed/tcga_brca_survival_clean.csv
# Output: figures/km_stage_plot.png
# =====

library(survival)
library(survminer)
library(tidyverse)
library(here)

df <- read_csv(here("data", "processed", "tcga_brca_survival_clean.csv"),
               show_col_types = FALSE)

# -----
# Kaplan-Meier estimator stratified by pathological stage
# -----
# survfit() computes the product-limit estimate of the survival function
#  $S(t) = P(T > t)$  within each stage group separately.
# Patients without an observed event by their last follow-up are censored:
# their exact survival time is unknown but contributes information up to that
# point.

fit <- survfit(Surv(os_time_months, os_event) ~ stage_group, data = df)

# -----
# Plot and export
# -----
# ggsurvplot() overlays all stage-group curves on one panel.
# risk.table = TRUE adds the at-risk count table below the plot, which is
# standard practice in clinical epidemiology publications.
# pval = TRUE displays the log-rank p-value testing the null hypothesis that
# all survival curves are identical.

png(here("figures", "km_stage_plot.png"), width = 1000, height = 800)
print(
  ggsurvplot(
    fit,
    data      = df,
    risk.table = TRUE,  # at-risk counts by time point
    pval      = TRUE,  # log-rank p-value
    xlab      = "Time since diagnosis (months)",
    ylab      = "Overall survival probability",
    legend.title = "Pathological stage",
    palette   = c("#2166ac", "#4dac26", "#d7191c", "#762a83")
  )
)
```

```
)  
dev.off()
```

```
cat("\nKaplan-Meier plot saved to figures/km_stage_plot.png\n")  
# Result: curves show progressively lower survival with advancing stage;  
# the log-rank test is statistically significant, motivating the Cox model  
# in 05_cox_regression.R.
```

5. Cox Regression and Diagnostics

Purpose: fit a Cox proportional hazards model, estimate hazard ratios, test the proportional hazards assumption, and create a forest plot.

```
# =====
# 05_cox_regression.R
# Cox proportional hazards regression – TCGA-BRCA overall survival
#
# Purpose:
#   Estimate the associations of age at diagnosis and pathological stage with
#   overall survival using a Cox proportional hazards model. Assess the
#   proportional hazards (PH) assumption via Schoenfeld residuals and
#   produce a forest plot of hazard ratios.
#
# Input:  data/processed/tcga_brca_survival_clean.csv
# Output: data/processed/cox_results.csv
#         data/processed/cox_ph_assumption_test.csv
#         figures/cox_forest_plot.png
# =====

library(survival)
library(broom)
library(tidyverse)
library(here)

# -----
# Load data and prepare model-ready dataset
# -----
df <- read_csv(
  here("data", "processed", "tcga_brca_survival_clean.csv"),
  show_col_types = FALSE
) |>
  filter(
    !is.na(stage_group),      # exclude patients with unknown stage
    !is.na(age),
    !is.na(os_event),
    !is.na(os_time_months),
    os_time_months > 0        # guard against zero follow-up (log undefined)
  ) |>
  mutate(
    # Set Stage I as the reference level so hazard ratios for Stages II-IV
    # are interpreted as relative to the lowest-risk stage group.
    stage_group = factor(
      stage_group,
      levels = c("Stage I", "Stage II", "Stage III", "Stage IV")
    )
  )

# -----
# Fit Cox proportional hazards model
# -----
# Surv() defines the time-to-event outcome: follow-up in months (continuous)
# and the binary event indicator (1 = death, 0 = censored).
# The model adjusts simultaneously for age (continuous) and stage (ordinal),
# so each hazard ratio is conditional on the other predictor.

cox_model <- coxph(
  Surv(os_time_months, os_event) ~ age + stage_group,
```

```

data = df
)

# tidy() extracts model coefficients; exponentiate = TRUE converts log-HR
# to HR scale for direct interpretation (HR = 1 means no association).
cox_results <- tidy(
  cox_model,
  exponentiate = TRUE,
  conf.int = TRUE
) |>
mutate(
  estimate = round(estimate, 3),
  conf.low = round(conf.low, 3),
  conf.high = round(conf.high, 3),
  p.value = signif(p.value, 3)
)

# -----
# Proportional hazards assumption test (Schoenfeld residuals)
# -----
# The Cox model assumes the HR is constant over follow-up time (PH assumption).
# cox.zph() tests this by regressing scaled Schoenfeld residuals on time.
# A significant p-value suggests the HR may vary over follow-up,
# in which case results should be interpreted as time-averaged associations.

ph_test <- cox.zph(cox_model)

ph_results <- as.data.frame(ph_test$table) |>
  rownames_to_column("term") |>
  as_tibble() |>
  rename(
    chisq = chisq,
    p_value = p
  ) |>
  mutate(
    chisq = round(chisq, 3),
    p_value = signif(p_value, 3)
  )

# -----
# Export model results
# -----
write_csv(cox_results, here("data", "processed", "cox_results.csv"))
write_csv(ph_results, here("data", "processed", "cox_ph_assumption_test.csv"))

cat("\nCox proportional hazards model results:\n")
print(cox_results)

cat("\nProportional hazards assumption test (Schoenfeld residuals):\n")
print(ph_results)

cat("\nCox model outputs saved to data/processed/\n")

# -----
# Forest plot of hazard ratios
# -----
# Each point is a HR estimate; horizontal bars are 95% CIs.
# The dashed vertical line at HR = 1 marks the null (no association).
# The x-axis is on the log scale so that symmetric CIs appear visually
# symmetric around the point estimate.

```

```

forest_data <- cox_results |>
  mutate(
    term_label = case_when(
      term == "age" ~ "Age, per year",
      term == "stage_groupStage II" ~ "Stage II vs Stage I",
      term == "stage_groupStage III" ~ "Stage III vs Stage I",
      term == "stage_groupStage IV" ~ "Stage IV vs Stage I",
      TRUE ~ term
    ),
    # Annotation text shown next to each point
    hr_label = paste0(estimate, " (", conf.low, "-", conf.high, ")")
  )

forest_plot <- ggplot(
  forest_data,
  aes(x = estimate, y = reorder(term_label, estimate))
) +
  geom_vline(xintercept = 1, linetype = "dashed", color = "gray40") +
  geom_errorbarh(
    aes(xmin = conf.low, xmax = conf.high),
    height = 0.18,
    linewidth = 0.8,
    color = "gray30"
  ) +
  geom_point(size = 3, color = "black") +
  scale_x_log10(
    breaks = c(1, 2, 5, 10, 20),
    limits = c(0.9, 30)
  ) +
  labs(
    title = "Cox proportional hazards model",
    subtitle = "Overall survival among TCGA-BRCA patients",
    x = "Hazard ratio (log scale)",
    y = NULL
  ) +
  theme_minimal(base_size = 13) +
  theme(
    plot.background = element_rect(fill = "white", color = NA),
    panel.background = element_rect(fill = "white", color = NA),
    text = element_text(color = "black"),
    axis.text = element_text(color = "black"),
    panel.grid.minor = element_blank()
  )

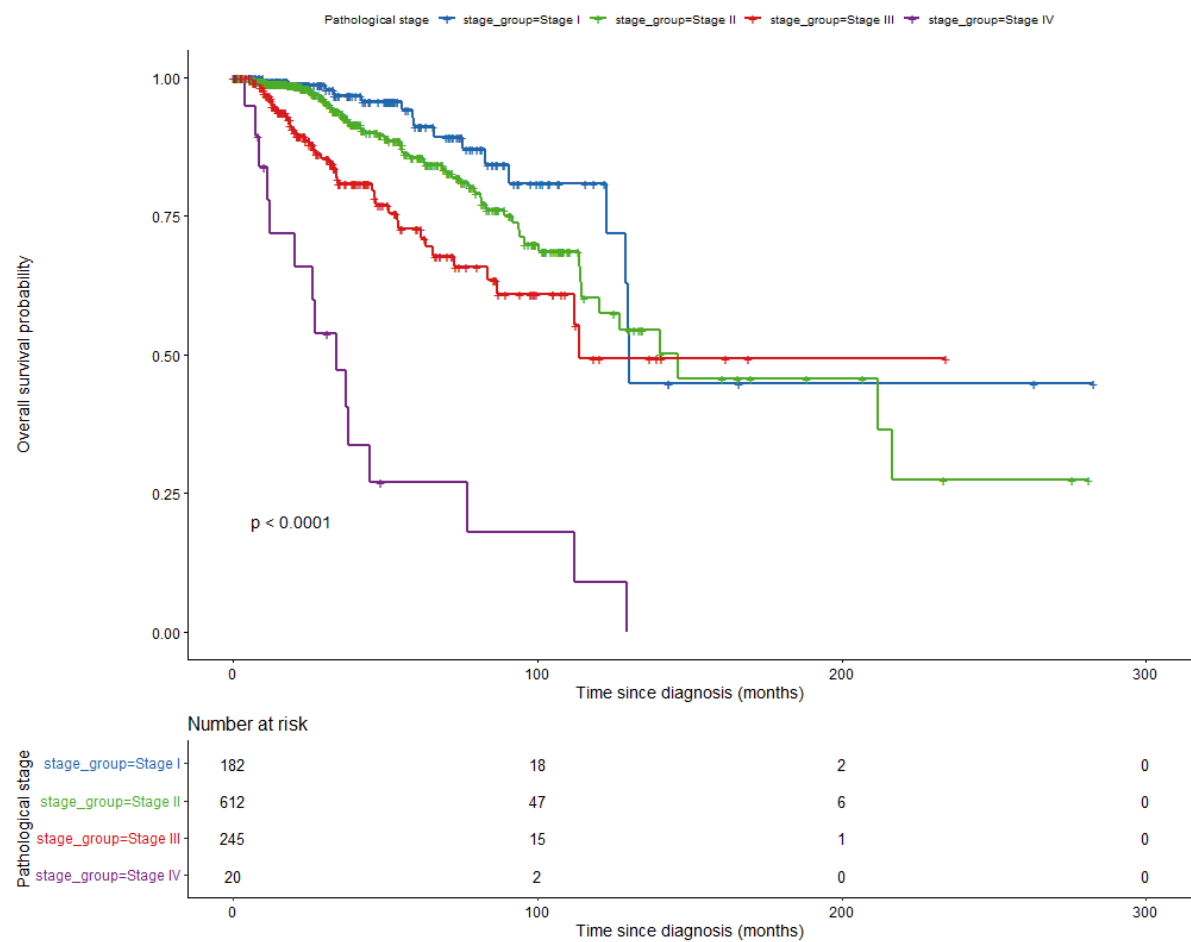
ggsave(
  here("figures", "cox_forest_plot.png"),
  forest_plot,
  width = 8,
  height = 5,
  dpi = 300,
  bg = "white"
)

cat("\nForest plot saved to figures/cox_forest_plot.png\n")

```

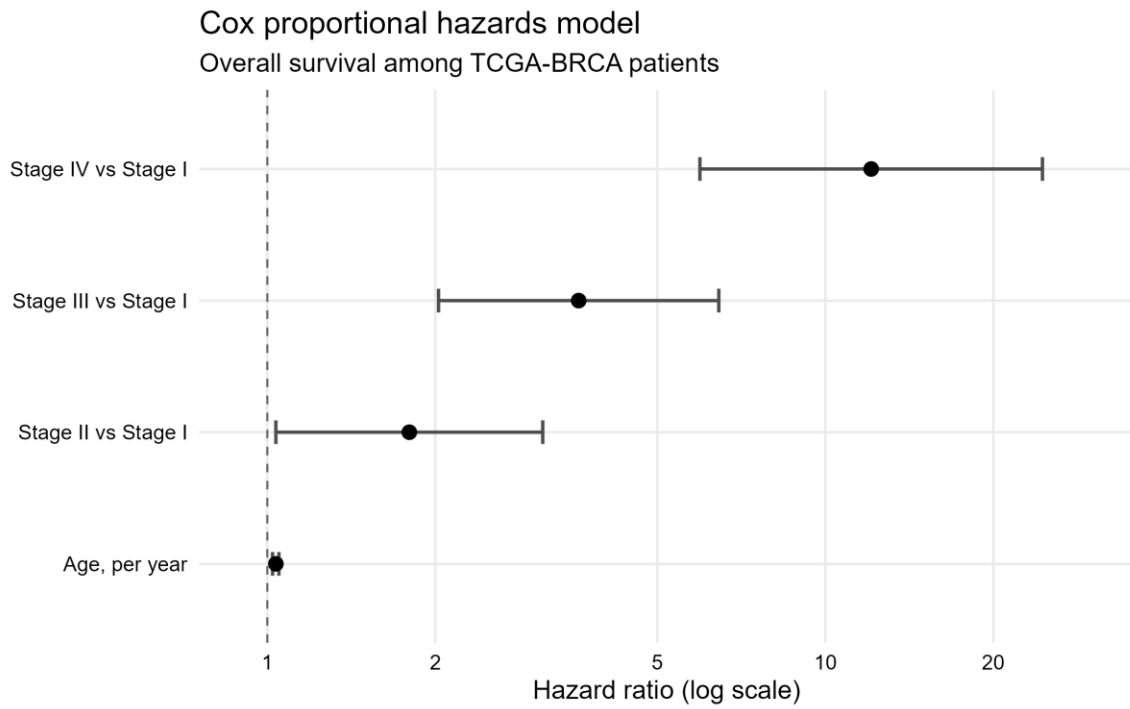
Figures

Kaplan-Meier Survival Curve



Kaplan–Meier curves show overall survival by pathological stage among TCGA-BRCA patients. More advanced stage is associated with lower survival, and the log-rank test is statistically significant.

Cox Model Forest Plot



Forest plot of Cox model hazard ratios adjusted for age and pathological stage. Stage I is the reference group; points represent hazard ratios and bars represent 95% confidence intervals.